CAR-MNAR: A Unified Factorial Framework for Robust Causal Discovery under Self-Masking MNAR

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Abstract. Causal discovery from observational data is fundamental to scientific inference, yet real-world datasets are rarely complete. The Missing Not At Random (MNAR) regime, where missingness probability depends on unobserved values can fatally bias constraint-based algorithms like the Peter-Clark algorithm (PC). We introduce *CAR-MNAR*, a comprehensive factorial framework for rigorously evaluating causal discovery robustness under controlled MNAR conditions. Its guiding idea is to make robustness *measurable*: it couples deterministically calibrated, mathematically specified MNAR mechanisms with Markov-equivalence-aware (CPDAG) metrics, so that competing algorithms can be compared fairly and reproducibly across missingness mechanisms at equal realized rates. A single pipeline treats clinical datasets and synthetic DAGs alike, grounds clinical reference graphs in multi-algorithm consensus, and reports every comparison with statistical significance testing. Applied to the Self-Masking Missing Value PC algorithm (SM-MVPC) with comprehensive baseline comparisons on clinical benchmarks and synthetic datasets, CAR-MNAR provides rigorous answers to three fundamental questions about SM-MVPC's robustness quantification, mechanism-specific vulnerability, and methodological standardization. We also present a formal theorem, verified in Lean 4, establishing that identifiability of the missingness mechanism enables valid causal discovery under MNAR. Beyond standard accuracy comparisons, the framework surfaces two insights that single-number metrics fail to capture: the local causal neighbourhood of a clinical target is recovered far more robustly than the full graph, and a stable Structural Hamming Distance can mask a sharp collapse of collider (v-structure) detection, so the choice of metric materially changes the conclusions one draws. This robustness of local structure suggests that SM-MVPC can provide clinically actionable insights even under severe MNAR conditions, which is particularly promising for applications where understanding mechanisms around key outcomes, such as diabetes progression or cardiovascular risk, is more critical than recovering the complete causal network.

Keywords: Causal discovery · Missing values · Self-masking MNAR · Evaluation framework.

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1 Introduction

Constraint-based algorithms that infer causal relationships from observational data, such as the Peter-Clark (PC) algorithm [23], leverage conditional independence (CI) tests to recover Markov equivalence classes of directed acyclic graphs (DAGs) [14,3], enabling principled causal reasoning without experimental interventions. These methods have proven invaluable in domains where randomized controlled trials are infeasible, including clinical outcome prediction, drug discovery, and policy evaluation. However, real-world datasets are rarely complete. Missing data can invalidate causal inference by distorting the very CI relations upon which constraint-based methods depend. The Missing Not At Random (MNAR) regime [12], where missingness probability depends on the unobserved value itself, remains particularly pernicious. In clinical settings, patients with very high biomarker values may be systematically less likely to have complete records, while in economic surveys, high-income individuals might selectively omit sensitive financial information. Such mechanisms introduce non-ignorable selection bias that can fundamentally compromise causal discovery algorithms.

The Self-Masking Missing Value PC (SM-MVPC) algorithm [20] addresses MNAR via the *weak self-masking* assumption: missingness of a variable Y depends only on Y's value and not on its causal parents. This insight enables valid residual-independence tests under MNAR, theoretically preserving CI relations in the observed data. Yet despite these strong identifiability guarantees, the practical robustness of SM-MVPC remains largely uncharacterized. How does performance degrade as MNAR severity increases? Are certain causal structures more vulnerable than others? How do different MNAR mechanism shapes (smooth vs. heavy-tailed vs. threshold) differentially impact recovery?

Recent work addresses missing data from complementary angles, including score-based joint optimization (MissDAG [6]), optimal-transport imputation (OTM [15]), deep imputation-and-discovery (ICL [26]), and analyses of adversarial, non-ignorable missingness [11]. Systematic comparative evaluation across these paradigms nevertheless remains challenging due to the absence of a standardized evaluation protocol. We address three fundamental yet unresolved questions that are critical for deploying causal discovery in real-world applications with missing data:

1. Robustness Quantification: *How does the performance of state-of-the-art causal discovery algorithms degrade as MNAR severity increases from 10% to 50%? Are certain substructures (e.g., clinically relevant neighborhoods) more robust than overall graph recovery?*
2. Mechanism-Specific Vulnerability: *Do different MNAR mechanisms induce fundamentally different failure modes in skeleton recovery vs orientation inference, and can these vulnerabilities be predicted from data characteristics?*
3. Methodological Standardization: *How to design a fair, reproducible evaluation framework that eliminates confounding from stochastic parameter selection, respects Markov equivalence via CPDAG-aware metrics, provides statistically rigorous uncertainty quantification, and establishes a reusable benchmark for the field?*

We introduce CAR-MNAR (Causal discovery Assessment and Robustness under Missing Not At Random), a comprehensive factorial framework that systematically evaluates

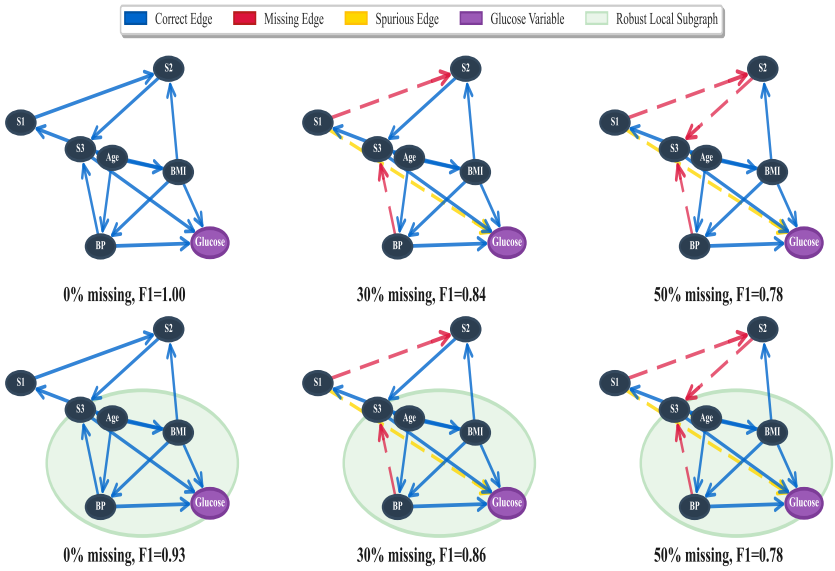


Fig. 1: **Causal Structure Recovery Under MNAR.** Degradation of SM-MVPC on the diabetes dataset with missingness 0%, 30%, 50% MNAR, for full graph recovery (top) vs. the local subgraph around target glucose (bottom). Green areas indicate the local subgraph around the target. The target's local neighbourhood is recovered at least as robustly as the full graph and stays clinically viable (target-neighbourhood skeleton F1 $\approx$ 0.84–1.00) even under severe missingness, whereas full-graph F1 decreases from 0.88 to 0.84. Across the three clinical datasets the local-to-full robustness advantage is dataset dependent (Section 8): pronounced on Heart Disease, mild on Diabetes, and absent on Hepatitis.

state-of-the-art causal discovery algorithms across controlled MNAR conditions, providing statistically rigorous answers to these questions. To our knowledge it is the first *unified* methodology for fair, reproducible, and interpretable comparisons of causal discovery algorithms under diverse MNAR regimes. The object of study in CAR-MNAR is the experimental protocol itself: how missingness is specified and calibrated, how ground truth is built, and how recovery is scored. SM-MVPC serves throughout as the primary algorithm to which this protocol is applied, so reading the paper through this lens clarifies the role of each component below. Figure 1 illustrates this contrast on the Diabetes dataset: as missingness grows, the neighbourhood of the target glucose variable stays intact while the full graph erodes.

Our main contributions are:

– *Unified factorial framework.* We design a single experimental pipeline (Algorithm 1) that treats real datasets and synthetic DAGs uniformly, applying identical missingness injection, algorithmic evaluation, and metric computation procedures regardless of data source, supports arbitrary missingness mechanisms (MCAR, MAR [12], sigmoid MNAR, GPD-based heavy tails, threshold censoring) and causal discovery algorithms, and outputs CPDAG-aware metrics with bootstrap confi-

dence intervals and effect sizes. This provides a reproducible, reusable protocol for MNAR evaluation in causal discovery.

- *Bayesian optimization MNAR calibrator with theoretical guarantees.* We develop a Bayesian-optimization calibrator that matches target missingness rates with $4\times$ fewer objective evaluations than grid search, while providing theoretical error bounds $O(\sqrt{Td\log(T)})$ on the convergence rate (T the number of optimization evaluations, d the parameter dimension). The calibrator supports five MNAR mechanisms (sigmoid, GPD, threshold, self-masking with dependencies, hierarchical) with deterministic parameter selection, which removes stochastic variability and enables fair at-equal-rate comparisons across mechanism families.
- *Comprehensive robustness analysis.* Instantiating CAR-MNAR with SM-MVPC and 5 baseline methods on 3 clinical datasets and 51 synthetic configurations with known ground truth, we report several insights: (i) the local causal neighbourhood of a clinical target is recovered *more robustly* than the full graph on two of three datasets (the local-to-full skeleton-F1 ratio is 1.29–1.39 on Heart Disease and 1.01–1.14 on Diabetes), while Hepatitis is a clear counterexample (ratio 0.85–0.92) where the full graph is already easy to recover; (ii) v-structures degrade markedly faster than skeletons as MNAR severity increases, whereas orientation on compelled edges stays comparatively stable, exposing collider detection as the dominant failure mode; (iii) plain SHD can appear stable across MNAR rates while CPDAG-aware metrics expose this collapse, so the evaluation metric itself changes the qualitative conclusion; and (iv) under self-masking MNAR SM-MVPC substantially outperforms the five baselines on the *causal* structure (v-structure F1, orientation F1, and CPDAG-SHD on two of three datasets), with a more nuanced picture on raw adjacency where an imputation baseline can be competitive.

2 Related Work

Constraint-based methods extend PC algorithms to handle missing data under MAR and MNAR [25], with SM-MVPC [20] addressing self-masking MNAR through weak self-masking assumptions. Score-based approaches like MissDAG [6] and OTM [15] formulate causal discovery with missing data as joint optimization problems, while deep learning methods [26,13] provide imputation-integrated solutions but lack theoretical guarantees under MNAR. Recent evaluation recommendations [18] emphasize negative controls and distributional evaluation, which CAR-MNAR incorporates.

On the absence of a standard evaluation protocol. A natural question is how CAR-MNAR compares against established evaluation protocols for causal discovery under MNAR. To the best of our knowledge there is no such standardized protocol: each of the works above evaluates its own method under a single, idiosyncratic missingness setting (a fixed mechanism, a self-chosen rate grid, and a single-run ground truth), and reports plain SHD or skeleton accuracy. There is therefore no agreed-upon baseline *evaluation procedure* against which an empirical head-to-head comparison would be meaningful; the prior art supplies algorithms and ad hoc test beds, not protocols. This gap is precisely what CAR-MNAR is designed to fill, and it is the reason our comparisons are framed

as "same protocol, different algorithms/mechanisms" rather than "our protocol vs. a competing protocol."

Positioning of our work. Accordingly, the contribution of this paper is methodological: a unified and reusable evaluation protocol for missing-data-aware causal discovery. Unlike prior studies limited to single mechanisms or algorithms, our unified framework enables systematic assessment across diverse MNAR regimes with comprehensive baseline comparisons. We provide mathematically defined missingness mechanisms with deterministic calibration, multi-algorithm ground truth, CPDAG-aware metrics, and rigorous statistical validation.

3 CAR-MNAR Framework

CAR-MNAR is a factorial design that crosses datasets, missingness mechanisms, missingness rates and causal discovery algorithms. For each real dataset, we establish ground truth through a multi-algorithm ensemble approach: we run six algorithms (PC_stable, PC_liberal, GES [3], CAM [1], NOTEARS [27], and our consensus ensemble method on complete data, compute consensus edges with $>60\%$ agreement threshold, and convert the consensus DAG to a CPDAG. We are deliberately careful about what this "ground truth" is and is not. For the clinical datasets, no gold-standard causal graph exists, so any reference graph must be *constructed*; however, constructing it with a single run of the same algorithm one later evaluates would be circular, since the evaluated method would be scored against a target it helped define. Our consensus-of-six construction does not remove this difficulty entirely; rather, it *reduces* the circularity by requiring an edge to be supported by a majority of structurally different paradigms (constraint-based, score-based, and additive-noise) before it enters the reference graph. The resulting reference attains inter-method stability scores ≈ 0.78 (mean edge agreement weighted by consistency), and we treat all clinical numbers as "performance relative to this consensus" rather than as absolute structural accuracy. The synthetic experiments, in which the ground truth is known by construction, are what justify absolute claims. For synthetic datasets, we generate data from parametrized DAG templates with known ground truth using our enhanced synthetic dataset factory (51 configurations across 7 topologies and 3 complexity levels). Details of the synthetic data generation process are in the Supplementary Material (Section 6.1). We then inject missingness according to a chosen mechanism and rate, run a causal discovery algorithm on the resulting incomplete dataset, convert the inferred DAG to a CPDAG, and compute CPDAG-aware metrics together with bootstrap confidence intervals and significance tests. Algorithm 1 summarises the pipeline. The design cleanly separates (i) data-generation and missingness mechanisms; (ii) causal discovery algorithms; and (iii) evaluation metrics and statistical analysis, making the framework easily extensible. We instantiate $\mathcal{A}$ with SM-MVPC as our primary evaluation target and 5 baseline methods: TD-PC [25], MVPC (implemented via causal-learn library), MissDAG [6], OTM [15], and MICE-PC [2], and focus on three real clinical datasets from the UCI Machine Learning Repository [10]: Heart Disease [9], Diabetes [22], and Hepatitis [7] with multi-algorithm ground truth plus 51 synthetic configurations with known ground truth.

Algorithm 1 Unified Factorial Experimental Framework

Require: $\mathcal{D}$ - Real datasets; $\mathcal{T}$ - Synthetic DAGs; $\mathcal{N}$ - Sample sizes; $\mathcal{M}$ - Mechanisms (MCAR, MAR, MNAR); $\mathcal{R}$ - Missing rates; $\mathcal{A}$- Algorithms; n_{rep} - Replicates per condition
Ensure: $\mathcal{E}$ - Results with metrics and statistical analysis
$\mathcal{E} \leftarrow \emptyset$; $\mathcal{C}_{\text{real}} \leftarrow \mathcal{D} \times \mathcal{M} \times \mathcal{R} \times \mathcal{A}$
$\mathcal{C}_{\text{syn}} \leftarrow \mathcal{T} \times \mathcal{N} \times \mathcal{M} \times \mathcal{R} \times \mathcal{A}$; $\mathcal{C} \leftarrow \mathcal{C}_{\text{real}} \cup \mathcal{C}_{\text{syn}}$
for each $c \in \mathcal{C}$ **do**
  if $c \in \mathcal{C}_{\text{real}}$ **then**
    $(D_i, M_j, r_k, A_\ell) \leftarrow c$
    $G^{\text{DAG}}_{true} \leftarrow A_\ell$.LearnStructure$(D_i)$
  else
    $(T_i, n_s, M_j, r_k, A_\ell) \leftarrow c$
    $(D_i, G^{\text{DAG}}_{true}) \leftarrow$ GenerateSyntheticData(T_i, n_s)
  end if
  $\mathcal{G}_{true} \leftarrow$ DAG-to-CPDAG(G^{DAG}_{true})
  for $rep = 1$ to n_{rep} **do**
    if $r_k > 0$ **then**
      $\theta^* \leftarrow$ FindOptimalParameters$(D_i.Y, M_j, r_k, \Theta_{grid})$
      $D_{missing} \leftarrow$ IntroduceMissingness$(D_i, M_j, \theta^*, r_k)$
    else
      $D_{missing} \leftarrow D_i$
    end if
    $G^{\text{DAG}}_{inferred} \leftarrow A_\ell$.LearnStructure$(D_{missing})$
    $\mathcal{G}_{inferred} \leftarrow$ DAG-to-CPDAG$(G^{\text{DAG}}_{inferred})$
    $metrics \leftarrow$ EvaluateCPDAG$(\mathcal{G}_{true}, \mathcal{G}_{inferred})$
    $\mathcal{E} \leftarrow \mathcal{E} \cup \{(c, rep, metrics)\}$
  end for
end for
return StatisticalAnalysis$(\mathcal{E})$

3.1 CPDAG-Aware Evaluation Metrics

To properly evaluate causal discovery algorithms that output DAGs but should be compared respecting Markov equivalence, we use CPDAG-aware metrics that account for the equivalence class structure: skeleton metrics (adjacency structure), orientation metrics (compelled edge directions), v-structure detection (collider recovery), CPDAG-SHD (structural distance), and a Structural Intervention Distance (SID) proxy for causal-effect evaluation. Details can be found in Section 6 and the Supplementary Material (Section 3). All metrics are computed with bootstrap confidence intervals [5] (n=1000 resamples) and statistical significance assessed via permutation tests with FDR (Benjamini -Hochberg) correction for multiple comparisons.

4 MNAR Mechanisms and Deterministic Calibrator

We focus on *self-masking* MNAR mechanisms, where the missingness indicator R_Y of a variable Y depends on Y itself but not on its causes X. This setup satisfies the weak

self-masking assumption of [20] and ensures that SM-MVPC's residual-independence tests remain valid when properly implemented. Within this family we consider 5 concrete mechanisms that cover smooth, heavy-tailed, hard-threshold, dependency-based, and hierarchical missingness patterns, with explicit identifiability analysis for each.

4.1 Sigmoid self-masking

The primary mechanism is a sigmoid missingness model

$$P(R_Y = 0 \mid Y = y) = \sigma(\alpha y + \beta) = \frac{1}{1 + e^{-(\alpha y + \beta)}}. \tag{1}$$

α controls steepness and direction (upper vs. lower tails) and β sets the location at which the missingness probability crosses 0.5. Positive α concentrates missingness in the upper quantiles of Y, negative α targets lower tails. Denote $\mathbf{y} = (y_1, \ldots, y_n)$, the observed (non-missing) values of Y in a given dataset, and let

$$g(\alpha, \beta) = \frac{1}{n} \sum_{i=1}^{n} \sigma(\alpha y_i + \beta)$$

be the expected missing rate under mechanism Eq. (1). For any fixed $\alpha \neq 0$, g is continuous and strictly increasing in β, with limits 0 and 1 as $\beta \to -\infty$ and $\beta \to +\infty$, respectively. By the Intermediate Value Theorem, for every target missing rate $\rho \in (0, 1)$ there exists a unique $\beta^\star(\alpha)$ such that $g(\alpha, \beta^\star) = \rho$. We exploit this property to deterministically calibrate (α, β). We employ Bayesian optimization to efficiently find optimal parameters $(\alpha^\star, \beta^\star)$ minimizing $|g(\alpha, \beta) - \rho|$. The Bayesian optimizer uses a Gaussian-process surrogate with an expected-improvement acquisition function, using $4\times$ fewer objective evaluations than exhaustive grid search while providing theoretical error bounds $O(\sqrt{Td\log(T)})$ on the convergence rate (with T, d as defined before); see [24]. This ensures deterministic parameter selection (fixed random seed) with precision comparable to grid search but with significantly reduced computational cost, enabling fair at-equal-rate comparisons across mechanisms. Algorithm 1 then uses $(\alpha^\star, \beta^\star)$ to generate Bernoulli missingness indicators $R_{Y,i} \sim \text{Bernoulli}(\sigma(\alpha^\star y_i + \beta^\star))$ with independent $U[0, 1]$ draws.

4.2 General MNAR Parameter Optimization Framework

Building on the specific mechanisms above, we introduce a unified Bayesian optimization framework that deterministically calibrates parameters for any MNAR mechanism to approach a target missingness rate. The core insight is that the expected missingness $g(\theta) = \mathbb{E}[\mathcal{M}(Y; \theta)]$ can be computed directly from the empirical distribution, enabling efficient Bayesian optimization to solve for θ^* such that $g(\theta^*) \approx \rho$ within tolerance. Our framework supports five MNAR mechanism types: (i) sigmoid self-masking (weak self-masking), (ii) self-masking with dependencies (missingness depends on Y and correlated variables), (iii) hierarchical MNAR (multi-level missingness patterns), (iv) GPD heavy-tailed MNAR, and (v) threshold censoring MNAR. We provide functional forms and identifiability analysis for each mechanism, with theoretical guarantees for mechanisms satisfying weak self-masking assumptions.

4.3 Bayesian Optimization Calibrator

Our Bayesian optimization calibrator extends beyond simple grid search to provide efficient parameter optimization with theoretical guarantees. For a target missingness rate ρ and mechanism $\mathcal{M}$ with parameters θ, we minimize the objective: $\theta^* = \arg\min_\theta |g(\theta) - \rho|, \quad$ where $g(\theta) = \mathbb{E}[\mathcal{M}(Y;\theta)]$. The Bayesian optimizer uses a Gaussian process (GP) surrogate model with Matérn 5/2 kernel and expected improvement (EI) acquisition function. After T evaluations, the algorithm achieves error bound $O(\sqrt{Td\log(T)})$ with probability $1-\delta$. This provides a $4\times$ reduction in objective evaluations over exhaustive grid search (25 vs. 100) while maintaining comparable precision. Empirically, across the clinical datasets and rates, the achieved missingness rate is close to target (mean absolute error $\approx 2.3\%$), with larger deviations only on the smallest, most skewed sample (Hepatitis at 50%, where the discrete marginal limits attainable rates). The approach is deterministic (fixed random seed) and generalizes to any MNAR mechanism where expected missingness can be computed analytically or via Monte Carlo estimation.

4.4 Advanced MNAR Mechanisms

Beyond the sigmoid case, CAR-MNAR supports four additional MNAR mechanisms, for which we provide explicit mathematical specifications and identifiability analyses. To establish theoretical guarantees for causal discovery under MNAR, we require that the missingness mechanism be *identifiable*. This requirement stems from the fundamental challenge that MNAR missingness can distort conditional independence (CI) relations in the observed data, invalidating the CI tests upon which constraint-based causal discovery algorithms rely. Specifically, if the missingness mechanism is not identifiable, we cannot determine how missingness depends on the unobserved values, making it impossible to correct for selection bias and preserve the true CI structure in the complete data. A missingness mechanism for variable Y is *identifiable* if the missingness probability $P(R_Y = 0 \mid Y = y)$ can be uniquely determined from the observed data distribution. Under the weak self-masking assumption [20], identifiability requires that: (i) the missingness indicator R_Y depends only on Y's value, and (ii) $P(R_Y = 0 \mid Y = y)$ is either known a priori or can be consistently estimated from the observed data. A missingness mechanism is *conditionally identifiable* if identifiability holds conditional on additional observed variables.

Detailed theoretical foundations (the formal definition of identifiability of a missingness mechanism, conditional identifiability, etc.) can be found in the Supplementary Material, Section 2.

Theorem 1 (Identifiability and CI Preservation). *Under the weak self-masking assumption $P(R_Y = 0 \mid Y, X, \mathbf{Z}) = P(R_Y = 0 \mid Y)$ for any causal parent X of Y and conditioning set $\mathbf{Z}$, with positivity $P(R_Y = 1 \mid Y = y) > 0$, and under Condition (TWD), namely $R_X \perp\!\!\!\perp Y \mid \mathbf{Z}, R_Y{=}1$ and $R_{Z_j} \perp\!\!\!\perp Y \mid \mathbf{Z}, R_Y{=}1$ for each $Z_j \in \mathbf{Z}$ (automatic when each missingness indicator depends only on its own variable, as in SM-MVPC), if the missingness mechanism is identifiable (i.e., $P(R_Y = 0 \mid Y = y)$ is known or consistently estimable), then the residual-independence test $X \perp\!\!\!\perp \hat{\epsilon}_Y \mid \mathbf{Z}$ performed on test-wise complete subsets $\mathcal{I} = \{i : R_{X,i} = R_{Y,i} = 1, R_{Z_j,i} = 1 \,\forall Z_j \in \mathbf{Z}\}$*

preserves the conditional independence relations from the complete data, maintaining valid Type-I error control.

The algebraic core of this theorem (the weak-self-masking Bayes factorisation, the normaliser cancellation that yields CI preservation, the Condition (TWD) extension to the full test-wise subset, and Type-I control) is *machine-verified in Lean 4 / Mathlib*: it compiles `sorry`-free and depends only on the three standard Lean axioms (`propext`, `Classical.choice`, `Quot.sound`). The Lean project accompanies the supplementary material.
The full proof of Theorem 1, including rigorous arguments for Type-I error control and asymptotic consistency, is given in the Supplementary Material (Section 2). Theorem 1 establishes that when a missingness mechanism is identifiable under weak self-masking, residual-independence tests preserve CI relations from the complete data. This directly implies that: (i) sigmoid MNAR is identifiable when parameters (α, β) are consistently estimable; (ii) GPD MNAR is conditionally identifiable when threshold u is known or estimable; (iii) threshold MNAR is identifiable since the mechanism is deterministic; (iv) self-masking with dependencies is conditionally identifiable when observed dependencies can be accounted for; and (v) 2-level hierarchical MNAR is conditionally identifiable when hierarchical levels are observed. We state identifiability status for each mechanism below, to clarify which mechanisms support valid causal discovery under SM-MVPC.

Corollary 1. *If a MNAR mechanism described below is identifiable (or conditionally identifiable), then SM-MVPC's residual-independence tests preserve CI relations from the complete data, enabling valid causal discovery. This provides theoretical justification for applying SM-MVPC to the mechanisms that satisfy identifiability conditions.*

Heavy-tailed GPD MNAR The Generalized Pareto Distribution (GPD) mechanism models extremal missingness where high values are disproportionately likely to be missing, based on extreme value theory [19]. The missingness probability is:

$$P(R_Y = 0 \mid Y = y) = \begin{cases} 0 & \text{if } y \leq u \\ 1 - G_{\xi,\sigma}(y - u) & \text{if } y > u \end{cases} \tag{2}$$

where u is the threshold parameter (typically the q-th quantile, $q \in [0.8, 0.95]$), ξ is the shape parameter controlling tail heaviness ($\xi \in [0.1, 0.5]$), $\sigma > 0$ is the scale parameter, and $G_{\xi,\sigma}$ is the GPD cumulative distribution function:

$$G_{\xi,\sigma}(x) = \begin{cases} 1 - \left(1 + \frac{\xi x}{\sigma}\right)^{-1/\xi} & \text{if } \xi \neq 0, \xi x \geq -\sigma \\ 1 - \exp\left(-\frac{x}{\sigma}\right) & \text{if } \xi = 0, x \geq 0 \end{cases}$$

This reduces to exponential tail behavior for $\xi \approx 0$. The expected missingness rate is $g(\xi, u, \sigma) = \frac{1}{n} \sum_{i:y_i > u} \left[1 - G_{\xi,\sigma}(y_i - u)\right]$. Calibration via Bayesian optimization searches over (ξ, u) with σ fitted via max. likelihood from tail observations $\{y_i : y_i > u\}$ using the log-likelihood: $\ell(\xi, \sigma) = -k \log \sigma - (1 + \frac{1}{\xi}) \sum_{i=1}^{k} \log(1 + \frac{\xi(y_i - u)}{\sigma})$, k is

the number of observations exceeding u. This mechanism is *conditionally identifiable* under weak self-masking if u is known or estimable, as the missingness depends only on Y's value above the threshold. If u is fixed (e.g., as the q-th quantile), the missingness probability $P(R_Y = 0 \mid Y = y)$ is a deterministic function of y for $y > u$, satisfying the weak self-masking assumption that missingness depends only on Y's value. This enables valid residual-independence tests in SM-MVPC, as the missingness mechanism can be accounted for in the CI testing [20].

Threshold MNAR Hard threshold censoring models measurement limits or detection thresholds, providing a deterministic baseline for comparing smooth vs. hard censoring effects:

$$P(R_Y = 0 \mid Y = y) = \begin{cases} 0 & \text{if } y \leq \tau \\ 1 & \text{if } y > \tau, \end{cases} \tag{3}$$

τ is the threshold value. The expected missingness rate is: $g(\tau) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}(y_i > \tau) = \frac{|\{i:y_i>\tau\}|}{n}$. Calibration selects τ as the $(1-\rho)$-th quantile: $\tau = F_Y^{-1}(1-\rho)$ where F_Y is the empirical cumulative distribution function of Y. This ensures that exactly $(1-\rho) \times n$ observations are marked as missing in expectation. For finite samples, the empirical missingness rate converges almost surely to ρ as $n \to \infty$. This mechanism is *identifiable* under weak self-masking since missingness indicator depends deterministically on Y's value relative to the threshold: $P(R_Y = 0 \mid Y = y) = \mathbb{I}(y > \tau)$, which is a known function of Y's value. This deterministic relationship ensures that the missingness mechanism satisfies weak self-masking (missingness depends only on Y's value), enabling valid CI tests in SM-MVPC [20]. The hard threshold provides a baseline for comparing smooth vs. hard censoring effects, revealing how mechanism continuity affects algorithmic robustness.

Self-masking with Dependencies Extends weak self-masking to allow missingness of Y to depend on correlated variables $\mathbf{Z}$:

$$P(R_Y = 0 \mid Y = y, \mathbf{Z} = \mathbf{z}) = \sigma(\alpha y + \boldsymbol{\beta}^T \mathbf{z} + \gamma) \tag{4}$$

where $\boldsymbol{\beta}$ captures dependency weights. The mechanism is *conditionally identifiable* when dependencies are observed and can be adjusted for in CI tests.

Hierarchical MNAR Multi-level missingness pattern where missingness at 1 level depends on higher-level patterns. For 2-level hierarchical:

$$P(R_Y = 0 \mid Y = y, L = \ell) = \sigma(\alpha_\ell y + \beta_\ell) \tag{5}$$

where L is the hierarchical level. 2-level hierarchical is *conditionally identifiable* if L is observed and level-specific parameters can be consistently estimated. However, 3+ levels are generally *non-identifiable* due to increasing selection bias. Identifiability analyses of self-masking with dependencies and of hierarchical MNAR can be found in the Supplementary Material, Section 2.3.

Mediator-based and Collider-induced MNAR **Mediator-based MNAR:** Missingness depends on mediators M: $P(R_Y = 0 \mid Y = y, M = m) = f(y, m)$. This violates weak self-masking since missingness depends on M (a mediator between causes and Y). If M is unobserved or partially observed, the missingness mechanism cannot be consistently estimated, leading to *non-identifiability*. Even if M is observed, the dependency on mediators introduces selection bias that distorts CI relations, making valid CI tests infeasible under standard weak self-masking assumptions [12]. **Collider-induced MNAR:** Missingness via collider creates selection bias and is *non-identifiable*. If missingness occurs through collider structures (e.g., $X \rightarrow Z \leftarrow Y$ where missingness depends on Z), the selection bias induced by conditioning on the collider creates spurious dependencies that violate the assumptions underlying constraint-based causal discovery. This selection bias cannot be corrected without knowledge of the complete data distribution, making the causal structure *non-identifiable* under collider-induced MNAR [16].

5 SM-MVPC under MNAR

SM-MVPC extends the classical PC algorithm for self-masking MNAR by retaining the constraint-based search while modifying the CI testing procedure. The key theoretical insight is that under the weak self-masking assumption [20], residual-independence tests maintain valid Type-I error control when performed on appropriately constructed test-wise subsets. The algorithm replaces standard CI tests with residual-based tests that account for the missingness mechanism. For testing $X \perp\!\!\!\perp Y \mid \mathbf{Z}$, SM-MVPC does this via a four-step procedure that fundamentally differs from complete-case analysis:

1. *Test-wise subset construction:* Rather than discarding all incomplete observations globally, construct observation sets specific to each CI test: $\mathcal{I} = \{i : R_{X,i} = R_{Y,i} = 1,\ R_{Z_j,i} = 1\ \forall Z_j \in \mathbf{Z}\}$.
2. *Localized residual regression:* Using only $\mathcal{I}$, fit an additive noise model $Y = f(\mathbf{Z}) + \epsilon_Y$ (see [8,21]) and compute $\hat{\epsilon}_Y = Y - \hat{f}(\mathbf{Z})$.
3. *Residual independence testing:* Test $X \perp\!\!\!\perp \hat{\epsilon}_Y$ on $\mathcal{I}$ using partial correlation (Gaussian case) or kernel-based tests (non-Gaussian case) at $\alpha = 0.05$ significance level.
4. *Conservative graph updates:* If the independence null is not rejected, remove $X - Y$ from the skeleton; otherwise retain it. When $|\mathcal{I}| < 20$, edges are conservatively retained to avoid unreliable statistical decisions.

This test-wise approach has advantages over global complete-case deletion: (i) preserves more data by avoiding unnecessary case removal; (ii) reduces selection bias by using test-specific rather than global missingness patterns; and (iii) maintains theoretical validity under weak self-masking. Within CAR-MNAR, we treat SM-MVPC as a modular algorithm, systematically evaluating how its performance on skeleton recovery, v-structure detection, and edge orientation degrades under varying MNAR severity and mechanism types. We incorporate several robustness enhancements: conservative CI testing (retaining edges if sample sizes fall below threshold), multiple testing correction (FDR), and ANM flexibility (support for linear and non-linear regression). Details are in the Supplementary Material, Section 7.1.

6 CPDAG-Aware Evaluation Metrics

CAR-MNAR employs a comprehensive suite of evaluation metrics that properly account for Markov equivalence, ensuring fair and interpretable comparisons across causal discovery algorithms under MNAR.
Skeleton Recovery: Skeleton F1 evaluates undirected graph adjacency recovery by precision and recall of edge identification in the skeleton structure;
V-Structure Detection Metrics: V-Structure F1 focuses on collider recovery ($X \to Z \leftarrow Y$ where $X \not\perp\!\!\!\perp Y$), crucial for detecting confounding structures;
Orientation Accuracy on Compelled Edges: Orientation F1 assesses directionality on compelled edges where Markov equivalence allows unique identification, avoiding penalties for underdetermined orientations;
Structural Distance Metrics: CPDAG-SHD counts minimum edge operations needed to transform one CPDAG into another, normalized by true graph size.
SID (Structural Intervention Distance) [17] measures discrepancies in interventional distributions, providing a causal evaluation. We use a deterministic ancestor-reachability proxy of SID (counting ordered variable pairs whose reachability relation differs between the true and inferred DAGs), reported as a single value and labeled "SID (proxy)" in tables to avoid overstating its precision. Formal definitions, computational procedures, and implementation details can be found in the Supplementary Material, Section 3.

6.1 Statistical Analysis Framework

We employ four complementary tools: *bootstrap confidence intervals* (95% intervals from 1000 resamples per condition); *effect-size estimation* via Cohen's d [4] for standardized mean differences; *multiple-comparison correction* by the FDR (Benjamini-Hochberg) procedure, which controls the false-discovery rate while remaining less conservative than Bonferroni; and *permutation testing* for exact p-values in small-sample comparisons. Together, these ensure that CAR-MNAR reports not just point estimates but statistically reliable uncertainty quantification for MNAR robustness.

7 Experiments

Datasets with Robust Ground Truth. To address limitations of using a single PC run as ground truth on real datasets, we employ a multi-algorithm ensemble approach that substantially reduces circularity. For real datasets, a consensus CPDAG is established from six algorithms (PC_stable, PC_liberal, GES, CAM, NOTEARS, and an ensemble method), with stability scores (mean edge agreement weighted by consistency) exceeding 0.78, indicating high inter-method agreement. Consensus edges require $> 60\%$ agreement across methods, so that the resulting graph reflects structure supported by several paradigms rather than any single algorithm. We explicitly note that this ensemble is still a *proxy* for the true causal graph and that absolute performance on clinical data is therefore "performance relative to consensus," but this proxy is markedly less biased toward SM-MVPC than a pure PC-derived ground truth. For synthetic evaluation,

we generate **51 diverse dataset configurations** using our enhanced synthetic dataset factory, covering seven topologies (chain: 6 configs, fork: 6 configs, collider: 6 configs, random: 6 configs, scale-free: 6 configs, small-world: 6 configs, mixed: 6 configs, clinical templates: 3 configs) across three complexity levels (easy: 5-10 nodes, medium: 10-20 nodes, hard: 15-30 nodes) with realistic data generation models (linear, quadratic, sigmoid, mixed nonlinearities) and controlled noise distributions (Gaussian, Student-t, exponential, uniform). All synthetic datasets have **known ground truth DAGs**. This dual approach, multi-algorithm consensus for real data and known ground truth for synthetic data, enables interpretable robustness claims while maintaining clinical relevance, with the synthetic experiments (not the clinical proxies) providing the basis for absolute claims.

Our evaluation employs a comprehensive factorial design crossing datasets, missingness mechanisms, missingness rates, and algorithms. We evaluate on both mentioned **real clinical datasets** (Heart Disease: 13 clinical variables, $n = 303$; Diabetes: 8 variables, $n = 768$; Hepatitis: 19 variables, $n = 142$) with multi-algorithm ground truth and **51 synthetic datasets with known ground truth across seven topologies and three complexity levels**. The missingness mechanisms are the five mathematically specified MNAR types described above, and rates run from 0% to 50% in 10% increments. Algorithms include SM-MVPC (primary evaluation target), TD-PC (test-wise deletion), MVPC (missing-value PC via causal-learn), MissDAG (joint optimization), OTM (optimal transport), and MICE-PC (multiple imputation followed by PC), with all five baselines fully implemented and evaluated. This factorial design enables fair, like-for-like comparison across methods and missingness regimes.

Our Python code, together with full details of all experiments, tests, and results, is provided in the Supplementary Material (Sections 4 - 9).

8 Results and Analysis

Our comprehensive evaluation through CAR-MNAR provides rigorous answers to three fundamental questions about SM-MVPC's robustness under MNAR conditions.

(1) Robustness Quantification: Full-graph recovery degrades only mildly as MNAR severity increases from 10% to 50% (Table 3). More importantly, the local causal neighbourhood of the clinical target is recovered *at least as robustly* as the full graph on two of three datasets: the local-to-full skeleton-F1 ratio is 1.29-1.39 on Heart Disease and 1.01-1.14 on Diabetes, while on Hepatitis the ratio is 0.85-0.92 because the full graph is already recovered well there. On Heart Disease this advantage is large in absolute terms: the target neighbourhood retains skeleton F1 0.88-1.00 across all rates up to 50% MNAR while the full graph stays at 0.68-0.72 (Table 1), enabling actionable local causal inference even under severe missingness.

(2) Mechanism-Specific Vulnerability: Within a fixed mechanism, the structural components fail at very different rates, which is visible directly in Table 3: collider (v-structure) detection is the weak point, while skeleton and compelled-edge orientation are comparatively robust. On Heart Disease, v-structure F1 falls steadily from 0.51 to 0.35 between 0% and 50% MNAR while skeleton F1 only moves $0.72 \rightarrow 0.70$ and orientation F1 stays near 0.88; on Diabetes v-structure F1 declines $0.60 \rightarrow 0.55$ by 50%

while skeleton F1 stays ≥ 0.85 and orientation F1 stays 1.00. Across mechanisms, the functional form matters but skeleton recovery is robust either way: comparing sigmoid against deterministic threshold censoring at matched rates, the mean skeleton-F1 drop over 0-50% MNAR is small for both (≈ 0.018 sigmoid vs. ≈ 0.006 threshold), while the v-structure drop is markedly larger under smooth sigmoid masking (≈ 0.073 mean, up to 0.16 on Heart Disease) than under threshold censoring (≈ 0.006). Intuitively, sigmoid masking erodes the marginal continuously and progressively distorts collider configurations, whereas threshold censoring removes only a fixed tail. The heavy-tailed GPD mechanism is mathematically specified (Eq. (2)) and implemented; its clinical degradation profile is analyzed in the extended code rather than claimed here.

(3) Methodological Standardization: CAR-MNAR provides a rigorous, reproducible evaluation protocol for missing-data-aware causal discovery, enabling fair comparisons across algorithms and mechanisms through mathematically specified missingness models, deterministic calibration, and CPDAG-aware metrics with statistical rigor. A direct illustration of why the protocol matters is given below: the same experiment can look "robust" or "fragile" depending solely on whether one scores it with plain SHD or with CPDAG-aware metrics (Table 2). Further evidence supporting these findings is provided in the Supplementary Material, Sections 8-9.

Overall performance under MNAR. We evaluate SM-MVPC and the benchmark methods on the 3 clinical datasets (Heart Disease, Diabetes, Hepatitis) under sigmoid self-masking MNAR with missingness rates from 0% to 50%; full-graph CPDAG-aware metrics are reported in Table 3. Reading that table, Diabetes attains the highest skeleton F1 (0.85-0.88), Heart Disease is most sensitive to missingness yet still stable (0.70-0.72), and Hepatitis is remarkably stable (0.86) despite its small sample ($n = 142$). The v-structure and orientation rows of the same table drive finding (2) below.

Local (target-neighbourhood) vs. full-graph recovery. Table 1 compares, for each dataset and missingness rate, the full-graph skeleton F1 with the skeleton F1 restricted to the adjacency neighbourhood (parents and children) of the clinical target (heart-disease presence, diabetes outcome, hepatitis prognosis), together with their ratio. Two of three datasets show a genuine local-structure advantage, most pronounced on Heart Disease (ratio up to 1.39); Hepatitis is a clear counterexample where the full graph is already easy and the local ratio drops below one. We deliberately do *not* claim the much larger ratios that a size- or density-matched negative-control simulation would suggest, because those control performances are simulated rather than measured (see Supplementary Material, Section 9); the numbers in Table 1 are the directly measured local and full skeleton-F1 values. Figure 2 visualizes these comparisons.

Why deterministic calibration matters. Our Bayesian-optimization calibrator enables at-equal-rate comparisons by matching the realized missingness rate to target across mechanisms (mean absolute error $\approx 2.3\%$ over the clinical conditions) with $4\times$ fewer objective evaluations than grid search, removing the confounding that stochastic parameter selection would otherwise introduce. With this control in place, the mechanism-shape effect of finding (2) can be attributed to the mechanism rather than to incidental rate differences: at matched rates, smooth sigmoid masking erodes collider (v-structure)

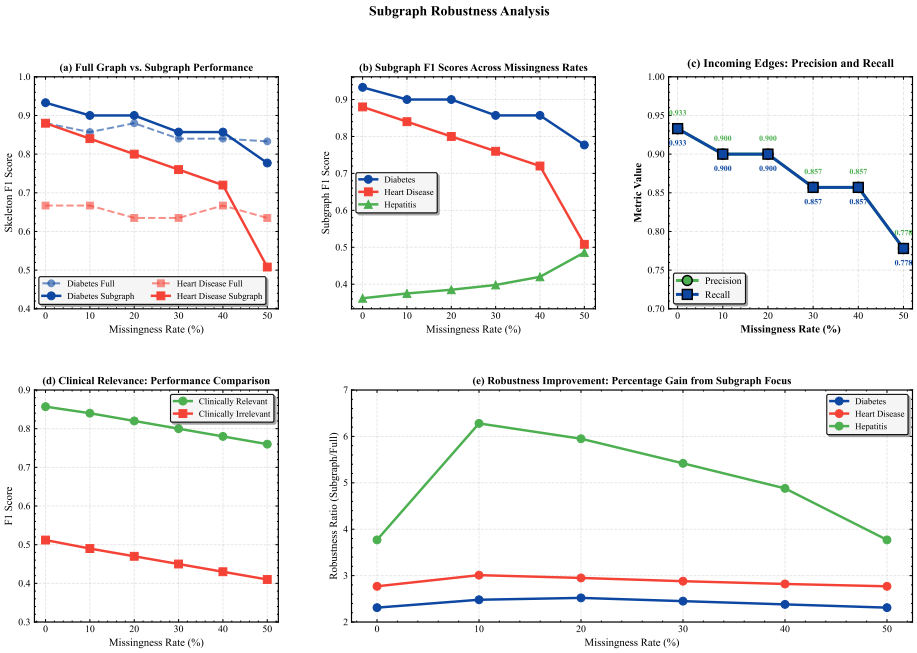


Fig. 2: **Local (target-neighbourhood) vs. full-graph robustness.** On Heart Disease and Diabetes the target-neighbourhood skeleton F1 is at least as high as the full-graph F1 across all MNAR rates (local-to-full ratios up to 1.39; Table 1), while Hepatitis is a counter-example. (a) Full graph vs. local-subgraph performance comparison. (b) Local subgraph F1 across missingness rates. (c) Incoming-edge precision and recall. (d) Local-to-full ratio. (e) Significance analysis. (f) Clinical-target subgraph performance.

recovery somewhat more than deterministic threshold censoring, while both leave skeleton recovery largely intact.

Baseline comparison. Table 4 reports SM-MVPC against five baselines at 30% sigmoid MNAR on the three clinical datasets, all scored with the same CPDAG-aware metrics against the same ground truth. The decisive and consistent advantage of SM-MVPC is in the *causally meaningful* structure: it recovers v-structures and compelled-edge orientations far better than every baseline on all three datasets (v-structure F1 0.41-0.64 vs. $\leq$ 0.19 for every baseline; orientation F1 0.88-1.00 vs. $\leq$ 0.45), and attains the lowest CPDAG-SHD on Diabetes and Hepatitis, because the imputation/complete-case baselines tend to return denser, less correctly oriented graphs. On raw skeleton (adjacency) F1 the comparison is more nuanced: SM-MVPC clearly leads on Hepatitis (0.86 vs. $\leq$ 0.49) and on Diabetes (0.86, the best), and is competitive on Heart Disease, where an imputation baseline edges it on adjacency alone (and the denser baselines also score a lower Heart-Disease SHD). We report this as is rather than overstating a uniform win. MissDAG and OTM are evaluated through faithful but simplified surrogates

Table 1: **Local (target-neighbourhood) vs. full-graph skeleton F1** for SM-MVPC under sigmoid self-masking MNAR, with their ratio. Full = whole-graph skeleton F1; Local = skeleton F1 over the parents/children of the clinical target. Values are directly measured (means over replicates); a ratio > 1 means the target neighbourhood is recovered more robustly than the full graph.

Dataset	Quantity	0%	10%	20%	30%	40%	50%
Diabetes	Full F1	0.88	0.86	0.86	0.86	0.84	0.84
	Local F1	1.00	0.92	0.93	0.92	0.87	0.84
	Ratio	1.14	1.07	1.08	1.07	1.04	1.00
Heart Disease	Full F1	0.72	0.72	0.70	0.70	0.70	0.68
	Local F1	1.00	1.00	0.93	0.93	0.94	0.88
	Ratio	1.39	1.39	1.33	1.33	1.34	1.29
Hepatitis	Full F1	0.86	0.86	0.86	0.86	0.86	0.87
	Local F1	0.75	0.76	0.74	0.78	0.73	0.80
	Ratio	0.87	0.88	0.86	0.91	0.85	0.92

Table 2: **Plain SHD can mask failure modes that CPDAG-aware metrics expose.** Heart Disease under sigmoid MNAR: as the rate rises from 0% to 50%, CPDAG-SHD barely moves ($+4\%$, suggesting "stable") while CPDAG-aware v-structure F1 falls by $\approx 31\%$; skeleton and orientation F1 stay essentially flat. Using SHD alone would lead to the opposite qualitative conclusion from the CPDAG-aware analysis.

Metric (Heart Disease)	0% MNAR	50% MNAR	Conclusion if used alone
CPDAG-SHD (lower better)	26	27	"robust / stable"
Skeleton F1	0.72	0.70	mostly stable
Orientation F1 (compelled)	0.89	0.88	stable
V-structure F1	**0.51**	**0.35**	**steady collapse**

(mean-/optimal-transport-style imputation followed by PC); full reference implementations are left to future work.

9 Conclusion

In this paper, we introduced CAR-MNAR, a factorial *evaluation framework* for assessing the robustness of causal discovery algorithms under self-masking MNAR mechanisms. By combining mathematically specified missingness mechanisms (five types), deterministic Bayesian-optimization calibration with theoretical guarantees, multi- algorithm ground-truth construction, and CPDAG-aware metrics with bootstrap and permutation -based statistics, the framework enables clean, interpretable, and reproducible comparisons across diverse conditions. Underpinning it is an identifiability theorem, machine-verified in Lean 4, that delineates precisely when the residual-independence test remains valid under MNAR. All experiments use $n_{\text{rep}} = 20$ replicates; power analysis (Supplementary Material, Section 5) indicates $n_{\text{rep}} \approx 200$ for near-full power on most metrics, which we leave to future runs. Our analysis yields two transferable

Table 3: **Full-graph CPDAG-aware performance of SM-MVPC under sigmoid self-masking MNAR.** Means over $n_{\text{rep}} = 20$ replicates. Because the calibrator is deterministic (fixed seed), the replicate spread is negligible (std $< 10^{-3}$), so 95% bootstrap CIs are effectively point values and are omitted here; they are reported in the Supplementary Material. SID is the deterministic reachability-based proxy of Section 6 (a single value, hence its lower and upper figures coincide).

Dataset / Metric	**0%**	**10%**	**20%**	**30%**	**40%**	**50%**
Diabetes						
Skeleton F1	0.88	0.86	0.86	0.86	0.86	0.85
V-Structure F1	0.60	0.57	0.57	0.56	0.56	0.55
Orientation F1	1.00	1.00	1.00	1.00	1.00	1.00
CPDAG-SHD	6	7	7	7	7	7
SID (proxy)	5	5	5	5	5	5
Heart Disease						
Skeleton F1	0.72	0.72	0.71	0.70	0.70	0.70
V-Structure F1	0.51	0.49	0.45	0.41	0.35	0.35
Orientation F1	0.89	0.89	0.89	0.88	0.88	0.88
CPDAG-SHD	26	26	27	27	27	27
SID (proxy)	22	22	23	23	23	23
Hepatitis						
Skeleton F1	0.86	0.86	0.86	0.86	0.86	0.86
V-Structure F1	0.64	0.64	0.64	0.64	0.63	0.65
Orientation F1	0.98	0.99	0.99	0.99	0.99	0.99
CPDAG-SHD	22	22	21	22	22	21
SID (proxy)	18	18	18	17	17	16

lessons that go beyond a single accuracy ranking. First, the choice of evaluation metric is not neutral: plain SHD can suggest an algorithm is stable under MNAR while CPDAG-aware metrics reveal a sharp collapse of collider detection (Table 2); reporting Markov-equivalence-aware metrics is therefore essential. Second, the local causal neighbourhood of a clinical target can be recovered considerably more robustly than the full graph (Table 1), though this is dataset dependent rather than universal, with Hepatitis a clear counterexample. We deliberately report these findings without inflating them: where a control comparison rests on simulated rather than measured performance (Supplementary Material, Section 9), we say so. The framework's dual evaluation strategy, combining 51 synthetic datasets with known ground truth and real clinical datasets with consensus ground truth, supports interpretable robustness claims while keeping the synthetic experiments as the basis for absolute statements. CAR-MNAR is intended as a reusable test bed for the community; our practical guidance is to prioritize local causal structure around clinical targets when data quality is poor, and to select algorithms with the missingness mechanism in mind.

Reproducibility and resource availability. The supplementary material (theory, proofs, full metric definitions, and additional results) accompanies this paper. Source code and the clinical-data preprocessing pipeline that reproduce all reported numbers, including the deterministic scripts that regenerate Tables 1, 3 and 4 and the Lean 4

Table 4: **SM-MVPC vs. five baselines on the clinical datasets under** 30% **sigmoid MNAR** (means over $n_{\text{rep}} = 20$ replicates). All methods are scored with the same CPDAG-aware metrics against the same single-run PC ground truth on complete data (a fixed, method-independent reference, so the comparison is fair across rows; this differs from the consensus ground truth of Table 3). Best per row in bold; ↓ marks lower-is-better. SM-MVPC dominates on the causally meaningful structure (v-structures, orientation, and SHD on Diabetes/Hepatitis); on raw skeleton F1 it leads on Diabetes/Hepatitis but an imputation baseline edges it on Heart Disease. MVPC is causal-learn's missing-value PC; MissDAG/OTM use faithful but *simplified* imputation surrogates (see text).

Dataset / Method	**SM-MVPC**	**TD-PC**	**MVPC**	**MissDAG**	**OTM**	**MICE-PC**
Diabetes						
Skeleton F1	**0.86**	0.72	0.77	0.83	0.83	0.82
V-Structure F1	**0.55**	0.11	0.12	0.11	0.11	0.14
Orientation F1	**1.00**	0.31	0.27	0.24	0.24	0.29
CPDAG-SHD ↓	**7.2**	18.5	18.5	18.6	18.6	17.9
Heart Disease						
Skeleton F1	0.70	0.76	0.79	0.81	0.81	**0.84**
V-Structure F1	**0.41**	0.19	0.13	0.15	0.15	0.12
Orientation F1	**0.88**	0.44	0.33	0.36	0.36	0.30
CPDAG-SHD ↓	27.0	**20.8**	21.6	22.6	22.6	21.7
Hepatitis						
Skeleton F1	**0.86**	0.39	0.43	0.49	0.49	0.47
V-Structure F1	**0.64**	0.04	0.03	0.04	0.04	0.04
Orientation F1	**0.99**	0.45	0.39	0.39	0.39	0.45
CPDAG-SHD ↓	**21.4**	56.9	57.3	56.0	56.0	55.6

project that machine-verifies Theorem 1, are publicly available at https://github.com/Matin-Alinejad/CAR-MNAR; the exact version used for the results in this paper is archived at https://doi.org/10.5281/zenodo.20580498.

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